WASP-11b

R-0707

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_241204` · created 2026-07-27T22:26:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460648.6444 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.052
Transit depth (Rp/Rs)^2	3.0 +/- 1.8 [%]
Orbital Inclination (inc)	88.4 +/- 1.68
Airmass coefficient 1 (a1)	0.49 +/- 0.33
Airmass coefficient 2 (a2)	0.93 +/- 0.91
Scatter in the residuals of the lightcurve fit is	76.59 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.097 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.052 vs 0.131 (deviation 0.64σ)
Duration fit vs published	0.097 d vs 0.1075 d (deviation 0.52σ)
Mid-time O–C	-7.2 min
Depth SNR	2.6
Residual scatter	76.59 %

Light curve

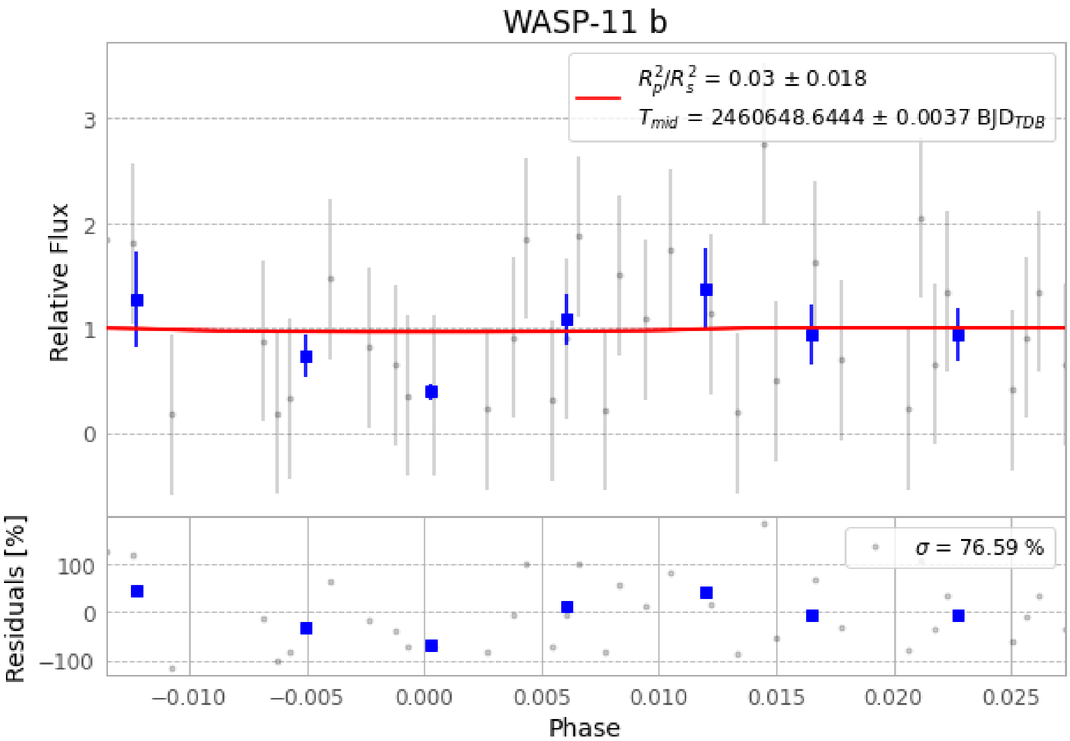

Figure 1 — FinalLightCurve_WASP-11 b_2024-12-03.png (SHA-256 be15be0276dce802…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [564, 379], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3be6a965c274986a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

74 FITS frames (49 MB), `HATP-10241204020616.FITS` → `HATP-10241204054515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-241204.log (72ca825f054a…), FinalLightCurve_WASP-11 b_2024-12-03.png (be15be0276dc…), FinalLightCurve_WASP-11 b_2024-12-03.csv (86dd8632c59d…)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)